

Fourier series.

For a function $f(x)$, which is piecewise smooth in the interval $-\pi \leq x \leq \pi$,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \quad (1)$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad (2)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad (3)$$

holds for all x where $f(x)$ is continuous. If $f(x)$ is discontinuous at $x = \xi$, then the Fourier series on the right-hand side of Eq. (1) converges to $\frac{1}{2}[f(\xi - 0) + f(\xi + 0)]$.

Examples

1. Expand in the Fourier series the following functions $f(x)$ defined in the interval $(-\pi, \pi)$:

$$(a) f(x) = \begin{cases} -\frac{\pi}{4}, & -\pi < x < 0, \\ \frac{\pi}{4}, & 0 < x < \pi, \end{cases} \quad (b) f(x) = \begin{cases} 0, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases} \quad (c) f(x) = |x|, \quad |x| \leq \pi.$$

In part (c), examine the answer for $x = 0$.

2. (a) Assuming that α is not an integer, find the Fourier series on $-\pi \leq x \leq \pi$ of the function $f(x) = \cos \alpha x$.
 (b) By setting $x = 0$ in the answer to part (a) show that for noninteger α

$$\frac{\pi}{\sin \pi \alpha} = \frac{1}{\alpha} + \sum_{m=1}^{\infty} (-1)^m \left(\frac{1}{\alpha + m} + \frac{1}{\alpha - m} \right).$$

- (c) By setting $x = \pi$ in the answer to part (a) show that for noninteger α

$$\pi \cot \pi \alpha = \frac{1}{\alpha} + \sum_{m=1}^{\infty} \left(\frac{1}{\alpha + m} + \frac{1}{\alpha - m} \right).$$

Using this formula for a particular value of α , find an expression for π .

Homework problems

1. Expand in the Fourier series the following functions $f(x)$ defined in the interval $(-\pi, \pi)$:

$$(a) f(x) = x, \quad -\pi < x < \pi, \quad [12]$$

$$(b) f(x) = x^2, \quad -\pi \leq x \leq \pi, \quad [13]$$

$$(c) f(x) = \sin \alpha x \text{ for } -\pi < x < \pi, \text{ assuming that } \alpha \text{ is not an integer.} \quad [10]$$

In (a) and (b) sketch the *periodic extensions* of the functions given by their Fourier series. Indicate the values taken by the Fourier series at discontinuities (if there are any).

2. Using the answer to question 1(b) for $x = 0$ and $x = \pi$, find the sums [10]

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \quad \text{and} \quad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

3. Show that the Fourier series for the function $f(x) = |\sin x|$ on $-\pi \leq x \leq \pi$ is

$$\frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2 - 1}.$$

Sketch the periodic extension of the function.

Using the above Fourier series for a particular value of x , deduce the value of [15]

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}.$$

4. Obtain the Fourier series for the function

$$f(x) = \begin{cases} 0, & -\pi \leq x \leq 0, \\ \sin x, & 0 \leq x \leq \pi. \end{cases}$$

Hence, deduce the value of [20]

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{4n^2 - 1}.$$

5. Obtain the half-range Fourier sine series of the function $f(x) = x(\pi - x)$ on $0 \leq x \leq \pi$.

[This is equivalent to assuming that $f(x)$ extends to $-\pi \leq x \leq 0$ as an odd function, i.e., $f(x) = x(\pi + x)$ here.]

To what value does the series converge for $x = \pi/2$? Deduce the value of the sum [20]

$$\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^3}.$$

$$\frac{x(1+uz)}{x(1+uz)} \sum_{\infty}^{0=u} \frac{u}{8} \cdot \frac{xu \cos \frac{zu}{n(1-)}}{z} \sum_{\infty}^{1=u} \frac{1}{z} + \frac{x}{z} \cdot \frac{xu \sin \frac{u}{1-u(1-)}}{z} \sum_{\infty}^{1=u} \frac{1}{z} \cdot \frac{1}{z}$$

Answers